

Agnes del Rocio Buscal Chang

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PROFESSIONAL SUMMARY

Industrial and Systems Engineer (Ranked 1st in class) merging Software Engineering, Applied Artificial Intelligence, and community leadership. Specialized in designing end-to-end solutions using Python, LLMs, and RAG architectures to solve environmental, scientific, and social challenges. Active technology evangelist, currently serving as an organizer and speaker for AWS User Groups and participating in the Aspire Leader Program. Highly motivated to foster technological innovation and drive the AI ecosystem forward in Latin America.

TECHNICAL SKILLS

- **AI / Machine Learning:** LLMs, NLP, RAG Architecture, Embeddings, Vector Databases, Generative AI APIs.
- **Programming Languages:** Python, JavaScript (Vanilla, React, Angular), Java, PHP, Swift (In Progress).
- **Cloud & Architecture:** Microsoft Azure, Azure Functions (Serverless), AWS, OAuth 2.0 Integration.
- **Data & Tools:** PostGIS, PocketBase, PostgreSQL, MySQL, Data Analysis.

SELECTED AI & ENGINEERING PROJECTS

GEO-HIDRO LLM | GEOTÓN PERÚ Contest *Python, React, PostGIS, RAG, Embeddings, Vector Database, LLM, Azure Functions*

- **Problem:** Analyzing pollution risks on national water networks required complex integration of geographic data and environmental variables.
- **Solution & Tech:** Developed an interactive platform integrating Geographic Information Systems (GIS) and AI. Utilized PostGIS for geospatial data management and implemented semantic search with a RAG architecture.
- **Impact:** The system dynamically identifies contamination sources within a 1km radius and generates contextualized risk evaluations and decision-making recommendations via LLMs.

BioSpace Navigator | NASA Space Apps Challenge (4th Place) *Python, Vanilla JS, LLMs, RAG, Embeddings, Vector Database, Azure Functions*

- **Problem:** Researchers struggled to efficiently extract specific insights from vast repositories of scientific literature.
- **Solution & Tech:** Led a development team to engineer a platform organizing over 600 NASA scientific documents. Built a semantic search engine and integrated a conversational LLM chatbot utilizing Generative AI APIs and RAG.
- **Impact:** Enabled users to query complex scientific data using natural language, receiving highly accurate and contextualized responses.

LIRA & LINO | AI for Social Impact *React, JavaScript, Tailwind CSS, PocketBase, RAG, Embeddings, REST APIs*

- **Problem:** Female artisans lacked a digital ecosystem to showcase their handmade products to a broader market.

- **Solution & Tech:** Created an end-to-end digital platform featuring an intelligent search engine based on semantic search and RAG capabilities, allowing users to discover unique artisanal pieces through natural queries.

PROFESSIONAL EXPERIENCE

Universidad de Piura | Piura, Peru

Junior Academic Applications Analyst | Sep 2025 – Present

- Developing an intelligent student survey processing system utilizing Python and AI (NLP/LLMs) to analyze qualitative feedback and generate automated teaching improvement recommendations.
- Architected "UDEP - IE", a comprehensive cloud platform for academic management, utilizing serverless Azure Functions and Microsoft OAuth 2.0 for high availability and strict data protection.
- Engineered advanced dynamic reporting and evaluation modules (Frontend and Backend) to digitalize institutional workflows for university stakeholders.
- Co-developing the foundation of a native game engine for Apple, leveraging SwiftUI and Metal to optimize graphical performance and user experience.

Web Development Intern | Feb 2025 – Aug 2025

- Participated in the integration of AI functionalities designed to provide intelligent recommendations within the centralized web system.
- Collaborated on the deployment and management of cloud infrastructure utilizing Microsoft Azure and Azure Functions (Serverless), ensuring data security via OAuth 2.0 authentication.

One Learning EIRL | Remote *Full Stack Developer* | Dec 2026 – Present

- Led the technological evolution of a business innovation platform, scaling the architecture from an initial landing page to a fully transactional B2C and B2B MVP.
- Constructed system architecture (PHP/MySQL) and integrated secure payment gateways (Niubiz) and robust SSO authentication modules.

LEADERSHIP & COMMUNITY INVOLVEMENT

- **Aspire Leader Program:** Active participant in the global leadership development program (In progress).
- **AWS User Group:** Organizer and Speaker, actively fostering the local cloud and tech community through events and knowledge sharing (In progress).
- **PRONABEC:** Volunteer Guide Talent (2023).

EDUCATION & CERTIFICATIONS

Universidad de Piura | Piura, Peru

Professional Title in Industrial and Systems Engineering | July 2026

Bachelor of Science in Industrial and Systems Engineering | July 2025

- **Honors:** Ranked 1st Place in the Graduating Class (Top 1%, 2020-2025).
- **Certifications in Progress:** AWS Certified Cloud Practitioner, Apple Swift Certification.

RELEVANT TRAINING & CONFERENCES

- **Google Research:** Regional Workshop on Data Science Applications and Research (2023).
- **HUB UDEP:** Future STEM Leaders (2024).
- **Universidad de Piura:** Data Analysis with Python Level 1, Fundamentals (2022).
- **Certiprof:** Scrum Foundation & Business Model Canvas Essentials (2024).